

Yash Mishra

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EDUCATION

Purdue University	West Lafayette, IN
Bachelor of Science in Computer Engineering	Aug 2023 – May 2026
Master of Science in Computer Engineering	Aug 2026 – Dec 2027
Cumulative GPA: 3.67/4.0; Dean's List and Semester Honors	
Relevant Coursework: Computer Architecture, Formal Verification, FPGA Design, ASIC Design, UVM Verification	

WORK EXPERIENCE

SoCET	West Lafayette, IN
UVM Verification Engineer	Aug 2024 – Apr 2025
- Architected modular UVM verification environments for an AMBA AHB-MUX IP, implementing drivers, monitors, sequencers, and predictors. - Designed coverage-driven verification strategy reducing pre-silicon escape defects by ~35%. - Developed TLM-based interfaces and constrained-random stimulus generation, cutting regression runtime by 25%. - Automated waveform debugging workflows and coverage reporting pipelines to accelerate root-cause analysis.	

NexSys Lab - Purdue University	West Lafayette, IN
Graduate Researcher	August 2026 – Present
- Developing read-locality-aware placement policies for ZNS SSDs within an LLM-assisted storage framework, extending a team prototype demonstrating 43.4% fewer garbage-collection copied bytes versus ZenFS in a RocksDB benchmark. - Investigating application access patterns and cross-zone data placement to reduce contention among concurrently accessed files while balancing read parallelism and garbage-collection efficiency. - Developing placement strategies for NVMeVirt-based evaluation, with planned deployment and hardware validation on the Cosmos+ FPGA-based OpenSSD platform.	

Teaching Assistant, Purdue University	West Lafayette, IN
ECE 270 (Digital Logic Design) & ECE 20875 (Python ML and Data Science)	Aug 2025 – Present
- Mentored 120+ students in digital logic design, Verilog-based RTL implementation, FSM design, and sequential circuit analysis, reinforcing theoretical concepts through lab problem-solving. - Led office hours for ECE 20875, explaining core machine learning concepts including Z-score normalization, stochastic gradient descent (SGD), loss functions, and statistical foundations, while guiding Python-based algorithm implementations.	

PROJECTS

PCIe Gen3 Endpoint Verification Environment SystemVerilog, UVM, SVA, Python, ModelSim	May 2026 – Present
- Developed a reusable UVM verification environment for a simplified PCIe endpoint, implementing drivers, monitors, sequencers. - Created 60+ constrained-random testcases and 12 functional covergroups, achieving 96% functional coverage across configuration, memory, and completion transactions. - Implemented 42 SystemVerilog Assertions (SVA) to verify protocol timing, packet ordering, flow control, and LTSSM transitions. - Automated regression execution and coverage reporting using Python, reducing verification turnaround by 30% while validating 10,000+ transactions per regression.	

FPGA-Implemented 5-Stage RISC-V Processor SystemVerilog, Vivado, ModelSim, FPGA Design	January 2026 - May 2026
- Designed an FPGA-based 5-stage pipelined RISC-V processor with hazard detection, forwarding, stall/flush control, and BTB. - Extended the design with private caches and MSI cache coherence, validating single- and dual-threaded benchmark workloads. - Developed 180+ directed and stress tests, achieving 100% ISA compliance through cycle-accurate simulation. - Synthesized and deployed on FPGA using Vivado, resolving timing violations and achieving up to 3.8× speedup over the baseline single-cycle design.	

SMT-Based Neural Network Formal Verification Engine Python, Z3, SMT, ONNX	April 2026 – July 2026
- Developed an SMT-backed verification engine that translated ONNX neural networks into Z3 constraints for formal verification. - Implemented 40+ SMT assertions to verify individual fairness, automatically generating counterexamples for violating models. - Evaluated the framework across 25 benchmark neural networks, enabling theorem-prover based verification without simulation.	

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, Python, C, C++, MATLAB
Verification: UVM, SVA, Constrained-Random Verification, Functional & Code Coverage, TLM, Formal Verification
Digital Design: RTL Design, RISC-V, Pipelined CPU Design, FSM Design, Computer Architecture, FPGA Implementation
Tools: ModelSim, Vivado, Quartus, Git, Linux
Embedded Systems & Hardware: STM32, ESP32, Zephyr RTOS, BLE, Sensor Integration, Interrupts